AIML 2nd Tom 22/04/24 M/S

USN | | | | | | | | | | [22TBSMA42]

FOURTH SEMESTER B.Tech EXAMINATION, APRIL-2024

MATHEMATICS & STATISTICS FOR MACHINE LEARNING

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**05 x02=10**
1.	**a.**	Define discrete random variable with a suitable example.	L-2	CO1	**02**
	b.	If the correlation coefficient between two variables is -0.75, what does it indicate about the variables?	L-2	CO3	**02**
	c.	The variance of a population is 2. The size of the sample collected from the population is 169. Find standard error of the distribution of means.	L-2	CO1	**02**
	d.	Define Null and Alternative hypothesis.	L-2	CO1	**02**
	e.	Find the Eigenvalues of the matrix $A = \begin{bmatrix} 4 & 0 \\ 3 & -5 \end{bmatrix}$.	L-2	CO6	**02**
		PART-B			**06 X 05=30**
2.	**a.**	A coin is tossed twice. A random variable X represents the number of heads. Find the discrete probability distribution of X. Also find the mean, variance and standard deviation.	L-3	CO2	**05**
		OR			
	b.	If 2% of electric bulbs manufactured by a certain company are defective, find the probability that in a sample of 200 bulbs i) less than 2 bulbs ii) more than 3 bulbs are defective.	L-3	CO2	**05**
3.	**a.**	The Ranks of 10 students in two subjects A and B are given below; the two numbers within brackets denote the rank of the same students in A and B respectively, (1,10), (2,9), (3,8), (4,7), (5,6), (6,5), (7,4), (8,3), (9,2), (10,1). Use spearman's Rank formula to find rank correlation coefficient.	L-3	CO3	**05**
		OR			
	b.	Find co-efficient of correlation for the data given below: X: 12, 15, 18, 21, 27 Y: 2, 4, 6, 8, 12	L-3	CO3	**05**
4.	**a.**	In a partially destroyed laboratory record of an analysis of correlation data the following results are given $\sigma_x^2 = 9$, Regression lines 8x − 10y + 66 = 0, 40x − 18y = 214. Find: **i.** Mean values of x and y **ii.** Coefficient of Correlation.	L-3	CO3	**05**

		OR			
	b.	Fit a straight line $y = ax + b$ for the following data.	L-3	CO3	**05**

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

5.	**a.**	For a particular brand TV picture tube, it is known that the mean operating life of the tubes is 1000 hrs with a standard deviation of 250 hrs, what is the probability that the mean for a random sample of size 25 will be between 950 and 1050 hrs? Given $[\emptyset(1) = 0.3413]$.	L-3	CO4	**05**
		OR			
	b.	The life time of a certain brand of an electric bulb may be considered as a random variable with mean 1200 hours and standard deviation 250 hours. Find the probability using central limit theorem that the average life time of 60 bulbs exceed 1250 hours. Given $[\emptyset(1.55) = 0.4394]$.	L-3	CO4	**05**
6.	**a.**	A correlation coefficient based on a sample of size 18 was computed to be 0.32. Can we conclude at 5% and 1% significance level that the corresponding population correlation coefficient is significantly greater than zero? Given $(t_{0.05,16} = 1.746, t_{0.025,16} = 2.120)$.	L-3	CO5	**05**
		OR			
	b.	An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed with mean of 800 hours and standard deviation of 40 hours. If a sample of 30 bulbs has an average life of 788 hours, use 5% level of significance to test the hypothesis H0: $\mu = 800$ hours against H1: $\mu \neq 800$. Also find the 95% confidence interval for the population mean of all bulbs produced by this firm. Given $(z_{0.025} = 1.96)$.	L-3	CO5	**05**
7.	**a.**	Find all the Eigen values and the corresponding Eigen vectors of the matrix $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$.	L-3	CO6	**05**
		OR			
	b.	Using diagonalization method, reduce the given matrix into diagonal form $A = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$.	L-3	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	20 wrist watches in a box of 100 are defective. If 10 watches are selected at random. Find the probability that **i.** 10 are defective **ii.** 10 are good **iii.** At least one watch is defective	L-3	CO2	**10**

		iv. At most 3 are defective.			
		OR			
	b.	The weight of ball bearings proved in a certain factory are normally distributed with mean 0.5 Newtons and standard deviation 0.002 Newtons. Find the percentage of ball bearings produced with weights, **i.** less than 0.498 Newtons **ii.** more than 0.501 Newtons and **iii.** between 0.4995 and 0.5005 Newtons. Given $[\emptyset(1) = 0.3413, \emptyset(0.5) = 0.1915,\ \emptyset(0.25) = 0.0987]$.	L-3	CO2	**10**
9.	**a.**	Find the regression of y on x for the following data. x: 10, 9, 8, 7, 6, 4, 3 y: 8, 12, 7, 10, 8, 9, 6	L-3	CO3	**10**
		OR			
	b.	The pressure P of a gas corresponding to various volumes V is recorded, and the data are given in the following. V cm^3: 50, 60, 70, 90, 100 P Kg/Cm^2: 64.7, 51.3, 40.5, 24.9, 7.8 The ideal gas law is given by the functional form $PV^{\gamma} = C$, where γ and C are constants. Estimate the constants C and γ.	L-3	CO3	**10**
10.	**a.**	Fit a second degree parabola in the least square sense to the following data. x: 1, 2, 3, 4, 5, 6, 7, 8, 9 y: 2, 6, 7, 8, 10, 11, 11, 10, 9	L-3	CO3	**10**
		OR			
	b.	Find the coefficient of correlation and the regression lines for the following data. x: 1, 2, 3, 4, 5 y: 2, 5, 3, 8, 7	L3	CO3	**10**
11.	**a.**	A population consists of five numbers 2,3,6,8,11. Consider all possible samples of size 2 which can be drawn with replacement from this population. Find (a) mean of the population (b) standard deviation of the population (c) mean of the sample distribution of means. (d) the standard error of sampling distribution of means.	L-3	CO4	**10**
		OR			
	b.	Explain Maximum Likelihood Estimation. Use maximum likelihood estimator method to find the parameters of Binomial distribution.	L-3	CO4	**10**

12.	**a.**	Fit a Poisson distribution for the following data. No. of mistakes in a page: 0, 1, 2, 3, 4, 5, 6 No. of pages: 275, 72, 30, 7, 5, 2, 1 Test the Goodness of fit at 5% level of significance. Given ($\chi^2_{0.05,6}$= 12.592).	L-3	CO5	**10**
		OR			
	b.	Define Type I and Type II errors. The heights of 10 males in a given locality are found to be 70, 67, 62, 68, 61, 68, 70, 64, 64, 66 inches. Is it reasonable to believe that the average height is greater than 64 inches? Test at 5% significance level assuming 9 degrees of freedom. Given (t = 1.833 at α =0.05).	L-3	CO5	**10**
13.	**a.**	Using diagonalization method, reduce the following matrix into diagonal form $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$.	L-3	CO6	**10**
		OR			
	b.	Given the following data, use Principal component analysis and find the covariant matrix. Feature: Example 1, Example 2, Example 3, Example 4 x: 4, 18, 13, 7 y: 11, 4, 5, 14	L-4	CO6	**10**